

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY
HOMEWORK 10

DUE: WEDNESDAY, APRIL 15, 2026, **BEFORE 23:59PM**

SUBMIT THE COMPLETED ASSIGNMENT TO <https://www.gradescope.com>

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- This assignment has **5 questions** on **two pages**.
 - This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
 - While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
 - Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
 - Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)
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1. Let

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

be a short exact sequence.

- (a) Prove that there exists a function $s : C \rightarrow B$ such that $g \circ s = \text{id}_C$, and that for any such s , the function $\phi : C \times A \rightarrow B$ given by

$$\phi(c, a) = f(a) + s(c)$$

is a bijection.

- (b) Prove that if $s : C \rightarrow B$ is a function satisfying $g \circ s = \text{id}_C$, then

$$s(c) + s(c') - s(c + c') \in \text{Ker}(g)$$

for all $c, c' \in C$, and deduce that

$$s(c) + s(c') - s(c + c') = f(\alpha_s(c, c'))$$

for a unique function $\alpha_s : C \times C \rightarrow A$.

2. *This question is a continuation of the previous question.*

- (a) Prove that the bijection ϕ becomes a homomorphism when the set $C \times A$ is given the binary operation $\diamond$ defined as

$$(c, a) \diamond (c', a') = (c + c', a + a' + \alpha_s(c, c')).$$

- (b) Consider the short exact sequence

$$0 \rightarrow \mathbb{Z} \xrightarrow{f} \mathbb{Z} \xrightarrow{g} \mathbb{Z}/10\mathbb{Z} \rightarrow 0,$$

with $f(x) = 10x$, $g(x) = [x]$. Let $s : \mathbb{Z}/10\mathbb{Z} \rightarrow \mathbb{Z}$ be the function which chooses the unique representative $x \in \{0, \dots, 9\}$ of $[x] \in \mathbb{Z}/10\mathbb{Z}$. Determine the function α_s in this case.

Remark, not part of the question: In a (very) early grade, perhaps you learned a method for adding natural numbers which involved “carrying”.

3. Let $A = (A_*, \partial^A)$ and $B = (B_*, \partial^B)$ be two chain complexes. For $p \in \mathbb{Z}$ we let

$$L_p = \prod_{n \in \mathbb{Z}} \text{Hom}(A_n, B_{n+p}).$$

In other words, L_p is the abelian group whose elements are tuples $\phi = (\phi_n)_{n \in \mathbb{Z}}$, consisting of a homomorphism

$$\phi_n : A_n \rightarrow B_{n+p},$$

for each $n \in \mathbb{Z}$. To such an element ϕ we associate $\partial^L \phi \in L_{p-1}$, defined by the formula

$$(\partial^L \phi)_n = \partial^B \circ \phi_n - (-1)^p \phi_{n-1} \circ \partial^A.$$

Prove that $\partial^L(\partial^L(\phi)) = 0$ for any $\phi \in L_p$ and any $p \in \mathbb{Z}$.

Remark: This makes $L = (L_, \partial^L)$ into a chain complex.*

4. Let $A = (A_*, \partial^A)$ and $B = (B_*, \partial^B)$ be two chain complexes, and let $L = (L_*, \partial^L)$ be the chain complex defined in the previous exercise.
- (a) Prove that $\phi \in L_0$ satisfies $\partial^L(\phi) = 0$ if and only if ϕ is a chain map.
 - (b) Let $\phi, \psi \in L_0$ satisfy $\partial^L(\phi) = 0 = \partial^L(\psi)$. Prove that $[\phi] = [\psi] \in H_0(L_*, \partial^L)$ if and only if ϕ and ψ are chain homotopic.

Remark: this identifies $H_0(L_, \partial^L)$ with the set of chain homotopy classes of chain maps $A \rightarrow B$. The chain complex L associated to A and B is sometimes denoted $\text{Hom}(A, B)$ or $\text{Hom}_*(A_*, B_*)$.*

5. Let X be a topological space and $A \subset X$ a subspace, and let $X \setminus A \subset X$ be the complement of A , equipped with the subspace topology. We then have a continuous bijection

$$k : A \sqcup (X \setminus A) \rightarrow X,$$

defined by $x \mapsto x$ for $x \in A$ and for $x \in X \setminus A$. Here, A and $X \setminus A$ are given the subspace topologies from X , and the domain of k is given the disjoint union topology: a subset is open if and only if its intersections with A and with $X \setminus A$ are open.

Assume in addition that X is path connected, and that $A \neq \emptyset$ and that $A \neq X$.

- (a) Show that the induced map

$$k_* : H_0(A \sqcup (X \setminus A)) \rightarrow H_0(X)$$

is not an isomorphism.

- (b) Deduce that the chain map in [Notes, equation (7.1) on page 67] does *not* induce an isomorphism on homology for $\mathcal{U} = \{A, X \setminus A\}$, when X and A are as in this question.

Remark (not part of the question): This shows that the assumption in [Notes, theorem 7.2.2] that $X = \bigcup \text{int}(U)$ cannot in general be weakened to $X = \bigcup U$.